

AHMED CHOWDHURY

SOFTWARE ENGINEER | BACKEND • CLOUD • DATA • AGENTIC AI

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SUMMARY

Software engineer building production backend, cloud, and data systems for enterprise applications, with experience spanning Java/Kotlin Spring Boot, GCP, MongoDB, Redis, BigQuery, and event-driven architectures. Recent work focuses on Python-based agentic AI and developer tooling using MCP, hybrid retrieval, and engineering automation.

TECHNICAL SKILLS

Languages: Java, Kotlin, Python, SQL

Backend & Data: Spring Boot, REST APIs, Microservices, MongoDB, PostgreSQL, Redis, SQL Server, BigQuery

Cloud & Distributed Systems: GCP, AWS, Pub/Sub, Kafka, Airflow, Cloud Run

AI & Agent Systems: MCP, RAG, BM25, Semantic Search, Hybrid Retrieval, Local Embeddings

DevOps & Observability: Terraform, Docker, Jenkins, GitHub Actions, CI/CD, Dynatrace

EXPERIENCE

Software Engineer | Ford Motor Company - Dearborn, MI

Apr 2025 - Present

- Independently designed and built a Python/MCP code-intelligence platform across 13 enterprise repositories using symbol-aware chunking, BM25 and local-embedding semantic retrieval, Reciprocal Rank Fusion ranking, and incremental content-hash indexing, cutting LLM context token consumption by ~30%.
- Built an agentic dependency-remediation workflow across 14 repositories that identifies FOSSA vulnerabilities, applies and validates fixes, and opens human-reviewed pull requests, reducing remediation time from ~4 hours to ~1 hour.
- Contributed to modernizing an enterprise order-processing system from SQL Server to MongoDB, consolidating regional order entities into a document model with compound indexes and collation, and wrote automated acceptance tests validating the migrated end-to-end flow, improving synchronization by ~15 minutes.
- Implemented reactive Redis caching with scheduled refresh, invalidation, and upstream API fallback under a ~100 request/second rate limit, removing per-request validation calls and improving request latency by ~400 ms.
- Build enterprise data pipelines moving operational CDC-style order data into BigQuery, where Python and Airflow workflows on Astronomer drive Terraform-managed BigQuery routines that transform it into analytical fact tables.
- Develop Java and Kotlin Spring Boot microservices and REST APIs across a 30+ service order-fulfillment platform with event-driven Pub/Sub flows, deployed through Terraform, Jenkins, and Cloud Run on GCP with Dynatrace monitoring.
- Contributed to an agentic migration workflow using persistent context, isolated worktrees, and local validation that produced 37 review-ready pull requests across 18 production Java microservices, cutting reported manual effort by over 90%.

Founding Full Stack Engineer | Perksway - Detroit, MI

Jan 2024 - Apr 2025

- Built and launched a K-12 rewards platform serving 500+ active users with Java, Spring Boot, React, and PostgreSQL, designing 15+ REST APIs and the data model behind authentication, roles, rewards, and redemptions.
- Containerized services with Docker and automated AWS deployments through GitHub Actions CI/CD.

Data Science & Engineering Intern | Ford Motor Company - Dearborn, MI

May 2023 - Aug 2023

- Built real-time Kafka and Spark Streaming pipelines feeding BigQuery analytics dashboards, reducing dashboard data population latency by ~25%.

PROJECTS

SafeSnap - AI-Powered Safety Incident Platform

Kotlin, Spring Boot, OpenAI API, Google Vision, PostgreSQL, AWS S3

- Incident platform with JWT/RBAC access control, presigned S3 uploads, asynchronous image processing, and AI-assisted root cause analysis gated by manager review.

EDUCATION

B.S. Information Technology, Business Administration Minor - Wayne State University | Dec 2024 | Summa Cum Laude | 3.93 GPA

CERTIFICATIONS

AWS Certified Solutions Architect - Associate | Jan 2025 - Jan 2028

Google Cloud Associate Cloud Engineer | Dec 2025 - Dec 2028